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Boosting brain tumor detection with an optimized ResNet and explainability via Grad-CAM and LIME

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Abstract
Detecting Brain Tumors is essential in medical imaging, as early and accurate diagnosis significantly improves treatment decisions and patient outcomes. Convolutional Neural Networks have demonstrated high efficiency in this domain, but their lack of interpretability remains a significant drawback for clinical adoption. This study explores the integration of Explainability techniques to enhance transparency in CNN-based classification and improve model performance through advanced optimization strategies. The primary research question addressed is how to improve the accuracy, generalization, and interpretability of CNNs for brain tumor Detection. While previous studies have demonstrated the effectiveness of deep learning for tumor detections, challenges such as class imbalance and overfitting of CNNs persist. To bridge this gap, we employ different dynamic learning rate modifiers, perform architectural enhancements, and apply XAI techniques, including Grad-CAM and LIME. Our experiments are conducted on three publicly available multiclass tumor datasets to ensure the generalizability of the proposed approach. Among the tested architectures, the enhanced ResNet model consistently outperformed others across all datasets, achieving the highest test accuracy, ranging from 99.36% to 99.65%. The techniques such as unfreezing layers, integrating various blocks, pooling, and dropout layers enhanced feature refinement and reduced overfitting. By incorporating XAI, we improve model interpretability, ensuring that clinically relevant regions in MRI scans are highlighted. These advancements contribute to highly reliable AI-assisted diagnostics, addressing significant challenges in medical image classification.

Keywords Brain tumor classification, Explainable AI (XAI), Convolutional Neural Networks (CNN), Grad-CAM

1 Introduction

Brain tumors are a complex class of illnesses that present significant challenges in detection and treatment [1]. According to the GLOBOCAN 2020 report, there were 308,102 new cases and 251,329 deaths related to the brain and central nervous system malignancies [2]. In India, data from the Hospital-Based Cancer Registry of the Indian Council of Medical Research indicates that brain tumors accounted for 1.6% of all cancer cases in 2021 [3]. Early and precise detection of these tumors is critical for improving patient outcomes. Magnetic Resonance Imaging (MRI) plays a central role in diagnosis, providing

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high-resolution anatomical information. However, manual MRI analysis is labor-intensive, subjective, and prone to human error [4], with inter-observer variability and cognitive fatigue further complicating accurate diagnosis [5]. The diverse and often rare pathology of brain tumors adds complexity, making early detection essential for determining appropriate treatment strategies. Delays in identifying tumor types can lead to poor outcomes, highlighting the need for reliable, interpretable, and efficient diagnostic tools. Deep learning (DL) and artificial intelligence (AI) have emerged as promising approaches to automate medical image interpretation [6]. Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in analyzing brain MRI images, capturing subtle patterns, and achieving high classification accuracy [7]. However, CNNs are often considered "black-box" models due to limited interpretability, which poses a critical concern in clinical applications. To bridge this gap, explainable AI techniques such as LIME (Local Interpretable Model-Agnostic Explanations) and Grad-CAM (Gradient-weighted Class Activation Mapping) has been employed to visualize regions contributing to model predictions [8]. Integrating Grad-CAM with enhanced ResNet architectures improves both the reliability and transparency of AI-driven brain tumor detection.

Recent studies have significantly advanced brain tumor detection, segmentation, and interpretability. Srinivasan et al. achieved 99.83% accuracy using channel-wise attention mechanisms with deep learning and five-fold cross-validation [9]. ResNet50 and InceptionV3 reached 99.34% and 97.70% accuracy for gliomas and pituitary tumors, respectively, through hyperparameter optimization [10]. Transfer learning with VGG-16 achieved 97.2% accuracy, though dataset diversity remained a limitation [11]. Ensemble techniques, such as GoogLeNet-based feature extraction [12], transformer-based models [13], and fuzzy logic with multilinear PCA [14], further enhanced classification. InceptionV4-based architecture attained 98.7% accuracy [15], while XGBoost-based ensembles emphasized preprocessing for tumor grading with 93% accuracy [16]. These studies collectively demonstrate the clinical potential of CNNs and hybrid models. Segmentation approaches have evolved alongside classification. Solanki et al. utilized PCA-based feature selection with attention mechanisms [17], whereas Alqhtani et al. combined diffusion filtering, fuzzy C-means clustering, and SVMs to achieve an accuracy of 0.982 with reduced processing time [18]. Morphological and unsupervised learning techniques also produced high precision [19]. Enhanced ResNet models improved segmentation over CNNs and FCNs by 10% [20], and multi-classification CNN frameworks achieved 95% recall [21]. Multi-modal approaches integrating deep neural networks with SVMs [22], ResNet50 with Grad-CAM [23], and YOLOv7-based pipelines [24] have further improved accuracy and interpretability. Hyperparameter-tuned ResNet50 models also demonstrated robust performance in tumor classification [25].

Beyond brain tumor detection, deep learning has been applied across biomedical domains. Neamah et al. reviewed tumor detection models, highlighting overfitting and the importance of transfer learning [26]. MobileNetV2L2 and CNN2 achieved 95–100% accuracy for lung diseases [27], while hybrid ensemble models with Random Forest and SMOTE reached 95.87% accuracy in coronary disease prediction [28]. Transformer-based models, such as BioBERT fine-tuned on PubMed [29], and CNN-SVM hybrid cancer models [30], expanded diagnostic capabilities. VGG16 achieved 94.5% accuracy in pneumonia detection [31], while feature engineering improved early dementia prediction [32] and SVMs enhanced early lung cancer detection [33]. EEG-based seizure detection benefited from feature selection [34, 35], and CNN modifications improved overall medical image classification [36]. Functional MRI has been employed for analyzing addiction-related brain networks [37], and enhanced residual networks reached 99% accuracy for MRI-based tumor classification [38]. Recent research also emphasizes explainability alongside accuracy. EfficientNetV2 with channel attention and Grad-CAM achieved 99.76% accuracy [39]. Hybrid CNN-ResNet frameworks improved efficiency and accuracy to 99.57% [40], while NeuroXAI integrated multiple explainability techniques for transparent tumor classification [41]. Image enhancement strategies, including Gaussian-blur sharpening, CLAHE, and DCT-based super-resolution with CNN-ResNet50, further improved classification performance, achieving 97.84–98.87% accuracy [42–44].Collectively, these advancements highlight the transformative potential of deep learning in brain tumor detection, segmentation, classification, and interpretability, establishing a solid foundation for clinically applicable AI-driven diagnostic systems.

Building on this foundation, the present study focuses on enhancing ResNet architectures to achieve superior accuracy while integrating explainability techniques, such as LIME and Grad-CAM, to provide interpretable insights into model predictions. By doing so, this work tries to address the gap between high-performance AI models and their clinical trustworthiness. The key contributions of this work are summarized as follows:

- An improved ResNet architecture is proposed by unfreezing the last ten layers, merging a squeeze-and-excitation block, adding global pooling, and integrating a dropout layer. These tunings have improved feature extraction, reduced overfitting, and enhanced multi-class classification accuracy.

- Analysis of detections with dynamic optimizers by adjusting the learning rate during training, mitigating the effects of class imbalance, and improving model generalization, particularly for the unseen tumor types.
- Integrating Grad-CAM and LIME to enhance interpretation by visualizing critical regions to improve model decisions. This contributes to clinically helpful AI-driven tumor detection models.
- Systematic evaluation of the results achieved for different architectural modifications and optimization strategies through an extensive ablation study, projecting their influence on classification performance.
- The proposed model, along with various deep learning-based detection mechanisms, is implemented and rigorously tested on three distinct brain tumor datasets to assess robustness and generalization.

The article's remaining sections are structured as follows: Sect. 2 elaborates on the methodology adopted in this work. Section 3 presents the explainable AI techniques used, including Grad-CAM and LIME, highlighting their role in interpreting model predictions. Section 4 reports the results and analysis of our research. Section 5 discusses the implications, strengths, and technical limitations of the proposed framework. Finally, Sect. 6 concludes the study and outlines future directions.

2 Methodology

The overall framework for brain tumor classification involves multiple stages, including preprocessing, model training, evaluation, and visualization of explainability. Initially, MRI images undergo resizing and one-hot encoding to standardize input for deep learning models. The proposed approach leverages an improved ResNet architecture with a squeeze-and-excitation block, global average pooling, dropout, and fine-tuning of the last ten layers to enhance feature extraction and classification performance. The model is trained and evaluated on multiple datasets, incorporating dynamic learning rate optimization techniques to improve generalization. Grad-CAM and LIME provide interpretability, highlighting the key regions influencing the model's predictions, and ensuring clinical reliability. Figure 1 is a visual description of the detailed system flow. Figure 2 is the pictorial description of the pipeline of the proposed work.

2.1 Dataset description

The brain tumor detection and classification dataset consists of MRI scans sourced from three publicly available datasets. These datasets provide grayscale MRI images in DICOM, JPEG, or PNG formats, with resolutions ranging from 256×256 to 512×512 pixels. The images are categorized into four main classes: glioma, meningioma, pituitary tumor, and no tumor, facilitating deep learning-based classification tasks.

Dataset 1 [45]: This dataset includes MRI scans split into training and testing sets. It contains 394 MRI images

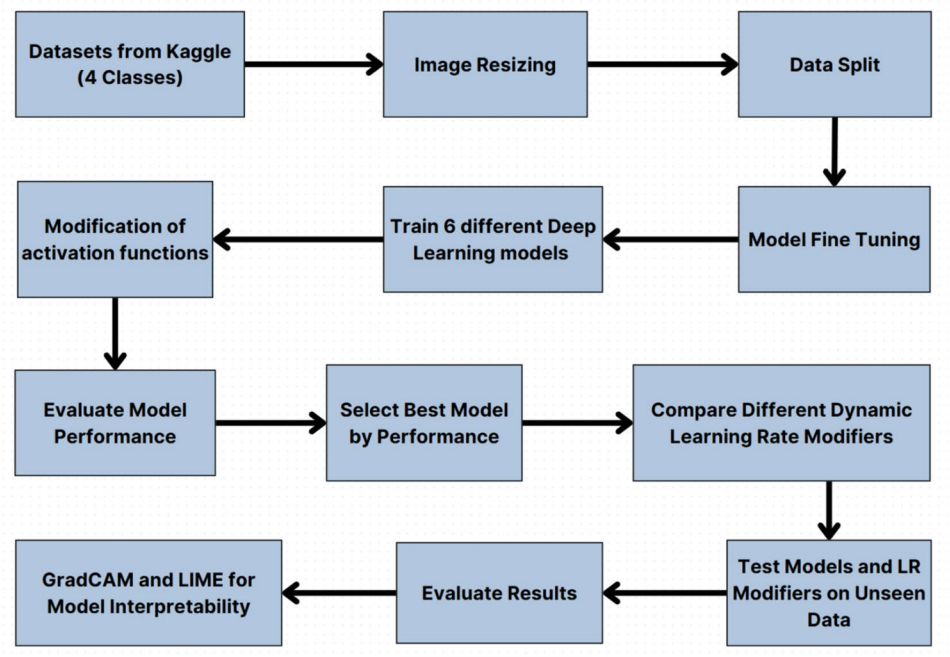

Fig. 1 System Architecture

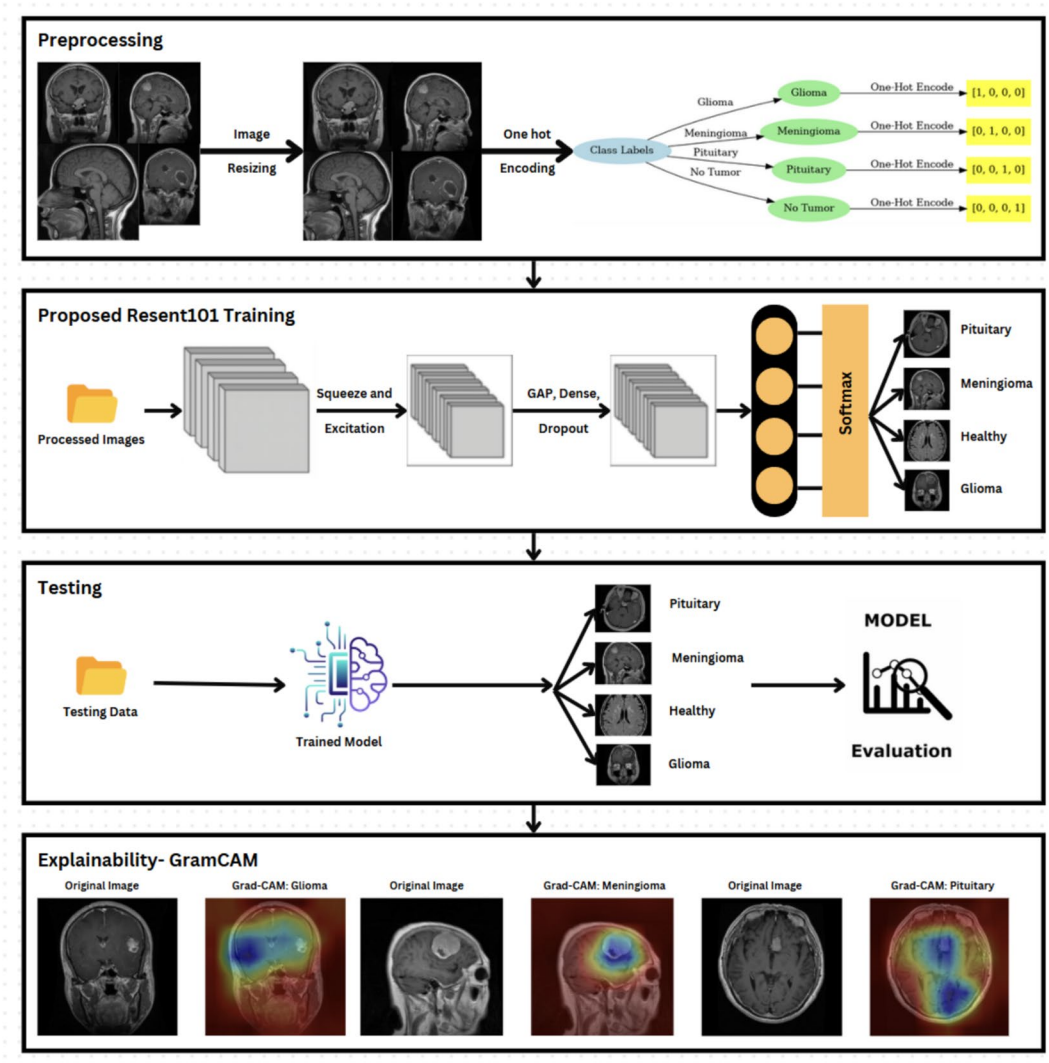

Fig. 2 Overall System flow Architecture

distributed across four tumor classes: glioma (100 images), meningioma (115 images), pituitary tumor (74 images), and no tumor (105 images).

Dataset 2 [46]: Comprising 7,023 MRI images, this dataset is structured into four categories: glioma (1,621 images), meningioma (1,645 images), pituitary tumor (1,757 images), and no tumor (2,000 images).

Dataset 3 [47]: A merged dataset combining three sources (Figshare, SARTAJ, and Br35H). It consists of 7,023 MRI images, divided into training and testing sets. The training set contains 5,712 images (glioma: 1,321, meningioma: 1,339, pituitary: 1,457, no tumor: 1,595), while the testing set has 1,311 images (glioma: 300, meningioma: 306, pituitary: 300, no tumor: 405).

The three detection-based datasets have been used, all of which are publicly available and licensed on Kaggle. These datasets contain brain tumor images categorized into four different classes of tumors. While they do not include patient data or expert annotations, they provide clear class labels for each image, making them suitable for training and evaluating object detection models. These datasets are specifically curated to facilitate tumor detection and classification tasks, supporting research in automated medical diagnosis.

2.2 Dataset preprocessing

In the dataset preprocessing phase, the MRI images are loaded from their respective directories based on labels such as 'glioma,' 'meningioma,' 'notumor,' and 'pituitary.' These images are resized to a consistent size of 250x250 pixels to ensure consistency before input into the model. The corresponding labels are then converted into numerical values through index mapping and subsequently one-hot encoded, which transforms the labels into a binary matrix format suitable for multi-class classification. The dataset is split into training and testing sets, with 80% of

the data allocated for training and the remaining 20% for testing, using the train_test_split method. Additionally, the pixel values of the images are normalized by dividing them by 255, ensuring that they are within the range of 0 to 1. This normalization helps speed up the model's convergence during training. These preprocessing steps are essential for preparing the data in the correct format for model training and evaluation.

2.3 Loss function and performance evaluation metrics

Loss Function for Multi-class Classification In multi-class classification tasks, the categorical cross-entropy loss is widely used to evaluate how well the predicted probability distribution aligns with the true class labels. The loss is computed as:

$$\mathcal{L} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c}) \tag{1}$$

where: N is the number of samples. C is the number of classes. $y_{i,c} \in \{0,1\}$ indicates the ground-truth label for class c of sample i. $\hat{y}_{i,c}$ is the predicted probability for class c of sample i.

The softmax function converts the model outputs (logits) into probabilities before applying this loss. This loss function encourages the model to assign higher probabilities to the correct class while minimizing the probability of incorrect predictions.

Performance Evaluation Metrics The effectiveness of the proposed model is assessed using multiple performance metrics evaluated for all the models, each highlighting different aspects of classification accuracy and reliability:

Training Loss (T-L, %): Measures the difference between the predicted and true values during training. Lower values indicate better convergence.

Training Accuracy (T-Acc, %): The proportion of correctly classified instances during training, calculated as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100 \tag{2}$$

Training Precision (T-Prec, %): Precision is the ratio of correctly predicted positive observations to the total predicted positives:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \times 100 \tag{3}$$

Training Recall (T-Rec, %): Recall is the ratio of correctly predicted positive observations to all actual positives:

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \times 100 \tag{4}$$

F1-Scores (G-F1, NT-F1, M-F1, P-F1): The harmonic mean of precision and recall for each class (Glioma (G), Non-Tumor (NT), Meningioma (M), and Pituitary Tumor (P)) is calculated as:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

2.4 Model training

In this research, several deep learning architectures with pre-trained models were utilized and trained to classify brain tumor types. Transfer learning was applied using pre-trained models such as MobileNetV2, EfficientNetV2-S, DenseNet-121, ResNet, and VGG-16. The convolutional base of each model was frozen since it had been trained on ImageNet to retain the learned feature representations. A custom CNN model is designed to contain multiple convolutional layers, batch normalization, pooling, and dense layers for extracting hierarchical features from the MRI scans.

The Adam optimizer with the categorical cross-entropy loss function, combined with metrics of accuracy, precision, and recall, was used to compile these models, as it may provide a fair measure of prognostic capability. Training was conducted on Colab Pro GPU NVIDIA A100 to accelerate computations, utilizing early stopping and learning rate reduction to achieve optimal performance. All the architecture performed well, as was evidenced by the efficacy of transfer learning and custom CNN approaches in the classification of brain tumors from MRI scans; most required fine-tuning: modifying the last unfreezing in a limited fashion by returning from 4 to 10 layers based on architecture into training at low learning rates. During training, the categorical cross-entropy loss was monitored using an Adam optimizer in the training environment, with accuracy, precision, and recall metrics employed to track its efficacy. Each model underwent training for 100 epochs, with a validation split incorporated and ran on a GPU, thus providing a thorough and systematic route to an effective multiclass brain tumor classification engine.

ResNet is selected as the best-performing model based on the results achieving high accuracy and efficiency. Transfer learning is employed, loading the pre-trained ResNet backbone with ImageNet weights and integrating its custom dense layers for classifying the images. Residual skip connections have enabled efficient training of extremely deep models by mitigating challenges such as the vanishing gradient problem. The framework architecture relies on several residual blocks composed of convolutional layers and skip connections, allowing for

the modeling of more complex pipeline processes out of the data, hence obtaining higher scores post-fine-tuning.

2.5 ResNet architecture and fine-tuning process

In Fig. 3, the proposed ResNet architecture is a powerful model for image classification tasks, leveraging residual connections to overcome the vanishing gradient problem. The fine-tuning of ResNet for specific tasks involves key components contributing to the model's ability to generalize effectively to new datasets [48]. The flow of the proposed methodology is illustrated in Fig.4. The architecture begins with an initial convolutional layer that applies a 7x7 convolution followed by batch normalization and a ReLU activation function. This layer extracts low-level features, such as edges and textures, laying the foundation for deeper layers to capture more complex patterns. The essential component of ResNet is the residual block, designed to facilitate gradient flow during

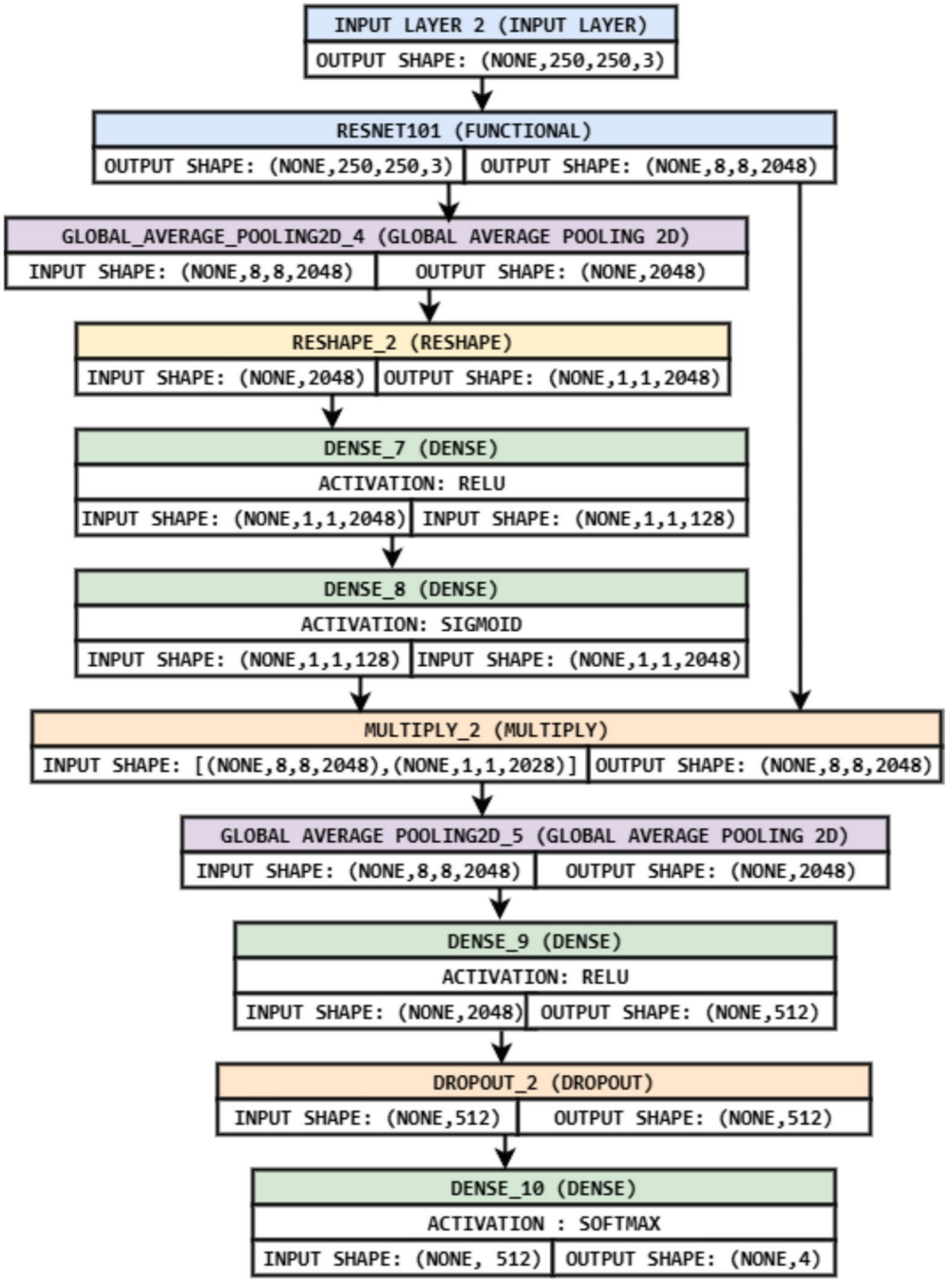

Fig. 3 Architecture of proposed Model

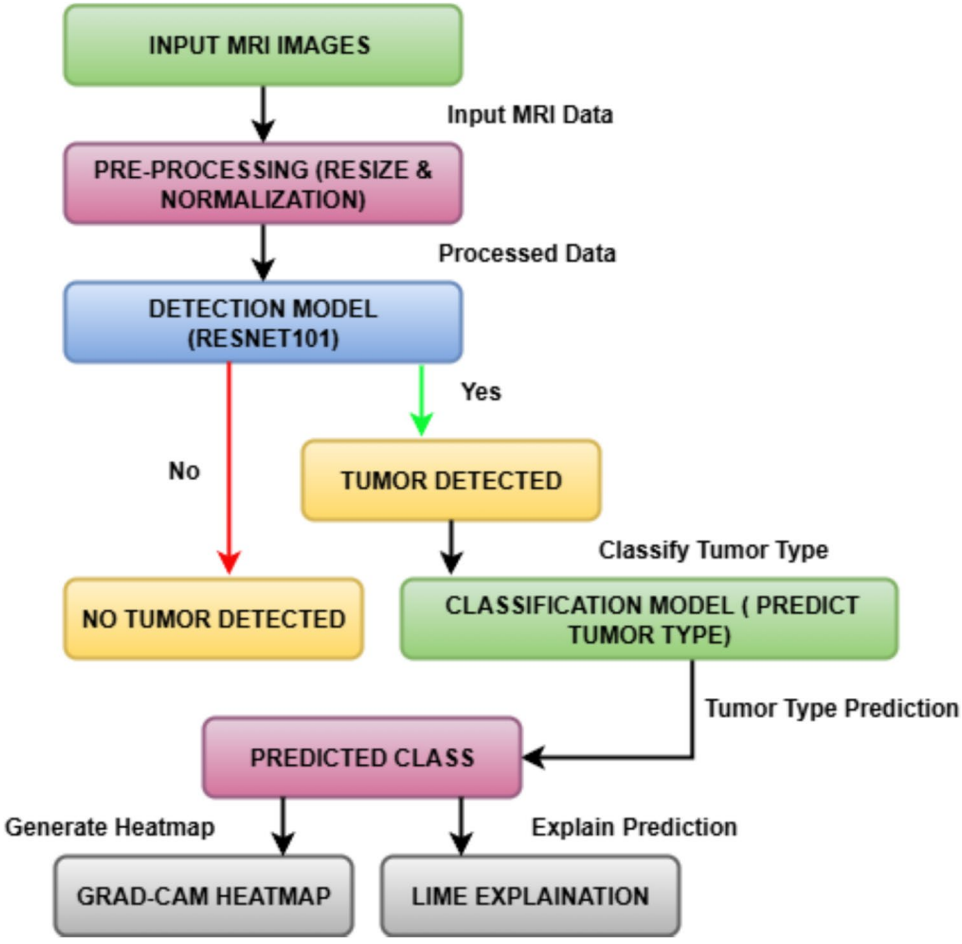

Fig. 4 Flow Diagram of Enhanced Brain Tumor Detection

backpropagation [49]. Each residual block is formulated as:

$$\text{Output} = F(x, \{W_i\}) + x \tag{6}$$

where $F(x, \{W_i\})$ represents the residual function – typically a stack of convolutional layers – and x denotes the input to the block. The addition of x to the residual function's output is termed a skip connection, which enables effective training of deep networks by preserving the flow of information. Following the residual blocks, the output feature map undergoes global average pooling (GAP), which condenses the spatial dimensions into a fixed-size feature vector [50]. GAP ensures that the model remains flexible to varying input image sizes while reducing parameters compared to traditional fully connected layers [51]. Finally, this feature vector is passed through a fully connected layer, typically followed by a softmax activation function, to produce class probabilities. This final layer is crucial to enabling the model to make predictions based on the high-level features extracted during earlier stages.

2.5.1 Squeeze-and-Excitation (SE) Block

In this study, we enhance the ResNet architecture by incorporating a Squeeze-and-Excitation (SE) block, which recalibrates channel-wise feature responses by explicitly modeling channel interdependencies [52]. The SE block consists of three primary operations: *squeeze*, *excitation*, and *scale*, which work together to enhance feature learning and improve model performance.

Squeeze Step

In the squeeze step, global average pooling is applied to each channel in the feature map to capture the global context. This results in a channel descriptor that summarizes the importance of each channel. Mathematically, the squeeze operation is represented as:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{ijc} \tag{7}$$

where z_c denotes the squeezed feature for channel c, H and W are the height and width of the feature map, and x_{ijc} is the feature at spatial location (i, j) in channel c.

Excitation Step

The excitation step learns channel-wise dependencies through two fully connected layers. The first layer reduces the dimensionality by a factor of r (reduction ratio), and the second layer restores the dimensionality to match the number of channels [53]. The excitation step is defined as:

$$s = \sigma\big(W_b \cdot \delta(W_a \cdot z)\big) \tag{8}$$

Here, W_a and W_b are weight matrices corresponding to fully connected layers, δ is the ReLU activation function, and σ denotes the sigmoid function, which generates the final scaling coefficients s.

Scale Step

Finally, the output of the excitation step is used to scale the input feature map, effectively adjusting the importance of each channel. The scaling operation is expressed as:

$$\hat{x}_{ijc} = x_{ijc} \cdot s_c \tag{9}$$

where s_c is the scaling factor for channel c, and $\hat{x}_{ijc}$ represents the modified feature map.

2.5.2 ResNet with SE block: implementation details

The SE-ResNet model enhances the standard ResNet architecture by integrating a Squeeze-and-Excitation (SE) block, improving channel-wise feature learning through recalibration. The Algorithmic representation is presented in Algorithm 1. The base model is a pre-trained ResNet without the top classification layers, initialized with ImageNet weights. The input image size is set to $250 \times 250 \times 3$. The SE block is incorporated after the final convolutional layer to emphasize significant feature channels. This SE block applies global average pooling (GAP) to compress the feature map $X \in \mathbb{R}^{H \times W \times C}$ into a channel descriptor $z \in \mathbb{R}^C$, where each channel descriptor is computed as:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{ijc} \tag{10}$$

The global average pooled vector is passed through two dense layers: the first reduces the feature dimension by a k factor, and the second expands it back to the original channel size. The attention weights are computed using:

$$a = \text{sigmoid}\big(U_2 \cdot \text{ReLU}(U_1 \cdot q)\big) \tag{11}$$

Algorithm 1 Proposed Improved ResNet-SE Architecture for Brain MRI Classification

Input: Input brain MRI image $\mathbf{X} \in \mathbb{R}^{250\times250\times3}$, filter size f, kernel size k, stride s, padding p, activation function a, dropout rate d_r
Output: Feature representation $\mathbf{O}$ with 4-class prediction
Initialization: Load pretrained ResNet weights (without top layers) and unfreeze last ten layers for fine-tuning.
Step 1: Residual Feature Extraction
For each residual block:
$\mathbf{F} \leftarrow \text{ConvBlock}(\mathbf{X}, f, k, s, p, a)$
$\mathbf{R} \leftarrow \mathbf{F} + \mathbf{X}$ Add skip connection
Step 2: Apply Squeeze-and-Excitation (SE) Block
$\mathbf{z} \leftarrow \text{GlobalAvgPool}(\mathbf{R})$
$\mathbf{s} \leftarrow \sigma(\text{Dense}(\text{ReLU}(\text{Dense}(\mathbf{z}))))$
$\mathbf{R}_{\text{scaled}} \leftarrow \mathbf{R} \odot \mathbf{s}$
Step 3: Global Feature Aggregation
$\mathbf{G} \leftarrow \text{GlobalAvgPool}(\mathbf{R}_{\text{scaled}})$
Step 4: Fully Connected Classification Head
$\mathbf{H} \leftarrow \text{Dense}(\mathbf{G}, 512, a)$
$\mathbf{H}_{\text{drop}} \leftarrow \text{Dropout}(\mathbf{H}, d_r)$
$\mathbf{O} \leftarrow \text{Dense}(\mathbf{H}_{\text{drop}}, 4, \text{softmax})$

Here, $U_1 \in \mathbb{R}^{C/k \times C}$ and $U_2 \in \mathbb{R}^{C \times C/k}$ are learnable parameters of the dense layers; ReLU introduces non-linearity, and $\mathrm{sigmoid}$ normalizes the output. The final recalibrated feature representation is obtained by scaling the original feature maps:

$$y_{i,j,c} = x_{i,j,c} \times a_c \tag{12}$$

Following the SE block, a GAP layer is applied to generate a fixed-size feature vector. This vector is then passed through a fully connected layer with 512 units and ReLU activation. Finally, the output layer contains 4 units corresponding to the number of classes, utilizing a softmax activation function to produce class probabilities. The model is optimized using the Adam optimizer with a learning rate of 0.0001 and categorical cross-entropy loss:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{13}$$

where C is the total number of classes, y_i is the true label, and $\hat{y}_i$ is the predicted probability for class i. Precision and recall metrics ensure effective learning and robust classification outcomes and enhance performance evaluation.

2.5.3 Dynamic learning rate modifiers

To further optimize model training, several dynamic learning rate modifiers were used:

Adam Optimizer

The Adam optimizer combines the benefits of Adagrad and RMSProp by using momentum and adaptive learning rates [54]. The update rules for Adam are as follows:

$$\begin{aligned} m_t &= b_1 \cdot m_{t-1} + (1 - b_1) \cdot g_t, \\ v_t &= b_2 \cdot v_{t-1} + (1 - b_2) \cdot g_t^2 \end{aligned} \tag{14}$$

$$\hat{m}_t = \frac{m_t}{1 - b_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - b_2^t} \tag{15}$$

$$\theta_t = \theta_{t-1} - \frac{lr \cdot \hat{m}_t}{\sqrt{\hat{v}_t} + eps} \tag{16}$$

Here, m_t and v_t denote the moving averages of gradients and squared gradients respectively, lr is the learning rate, b_1 and b_2 are decay factors controlling the weight of past estimates, and eps is a small constant to avoid division by zero [55].

RMSProp Optimizer

RMSProp dynamically adapts the learning rate by considering the moving average of the squared gradients [56, 57]. The update equations are given below:

$$v_t = b \cdot v_{t-1} + (1 - b) \cdot g_t^2 \tag{17}$$

$$\theta_t = \theta_{t-1} - \frac{lr \cdot g_t}{\sqrt{v_t + eps}} \tag{18}$$

Here, v_t represents the exponential moving average of squared gradients, g_t is the gradient at time step t, lr is the learning rate, and eps is a small constant used to prevent division by zero.

AMSGrad Optimizer

AMSGrad enhances the Adam optimizer by ensuring that the effective learning rate remains constant, thereby improving stability [58]. The update rules for AMSGrad are as follows:

$$v_t = b_2 \cdot v_{t-1} + (1 - b_2) \cdot g_t^2 \tag{19}$$

$$\hat{v}_t = \max(\hat{v}_{t-1}, v_t) \tag{20}$$

$$\theta_t = \theta_{t-1} - \frac{lr \cdot g_t}{\sqrt{\hat{v}_t} + eps} \tag{21}$$

In these equations, b_2 is the decay rate for the second moment estimates, v_t is the moving average of squared gradients, and $\hat{v}_t$ stores the maximum of past squared gradient estimates to avoid the learning rate from increasing.

ReduceLROnPlateau Scheduler

The ReduceLROnPlateau scheduler reduces the learning rate when the model's validation loss stops improving [59]. This helps the model converge more efficiently by decreasing the step size once progress slows down. The update rule is defined as:

$$lr_t = lr_{t-1} \times \mathrm{factor} \tag{22}$$

Here, lr_t represents the adjusted learning rate at step t, and the factor is a multiplicative constant that reduces the learning rate after the model performance plateaus [60].

3 Explainable AI using Grad-CAM and LIME

In medical imaging, understanding model decisions is crucial, particularly when dealing with critical tasks such as brain tumor detection. To enhance the interpretability of our optimized ResNet model, we employ two prominent Explainable AI (XAI) techniques: Gradient-weighted Class Activation Mapping (Grad-CAM) and Local Interpretable Model-agnostic Explanations (LIME). These techniques provide insights into the model's decision-making process by highlighting the regions in the input image that contributed most to the model's prediction.

3.1 Grad-CAM: Gradient-weighted class activation mapping

Gradient-weighted Class Activation Mapping (Grad-CAM) is a popular technique for visualizing the class-specific discriminative regions in an image. It produces heatmaps that emphasize the important areas contributing to a model's prediction, thus improving interpretability. For a trained CNN model, let $f^k(\mathrm{x})$ represent the predicted score for class k for an input image x. Grad-CAM computes the gradients of $f^k(\mathrm{x})$ with respect to the feature maps A^l from the final convolutional layer.

The importance of weight coefficients w_c^l (previously denoted as α_c^l) are computed as:

$$w_c^l = \frac{1}{Z} \sum_{i,j} \frac{\partial f^k}{\partial A_{ij}^l} \tag{23}$$

Here, Z denotes the total number of spatial locations in the feature map, and A_{ij}^l is the activation at location (i, j) of layer l. Next, these weights are used to compute the weighted sum of the feature maps as:

$$\hat{A}^l = \mathrm{ReLU}\left(\sum_c w_c^l A^l\right) \tag{24}$$

ReLU activation ensures that only positive contributions to the class score are considered, ignoring negative influences. Finally, the weighted feature map is resized to the same dimensions as the input image to create the Grad-CAM heatmap:

$$\mathrm{Heatmap} = \mathrm{ReLU}\left(\sum_l w_c^l A^l\right) \tag{25}$$

This heatmap visually highlights the regions of the input that most significantly impact the model's decision with better transparency and interpretability in deep learning-based medical diagnosis [61].

3.2 LIME: Local interpretable model-agnostic explanations

Local Interpretable Model-agnostic Explanations (LIME) is designed to explain predictions made by any black-box machine learning model. Unlike Grad-CAM, LIME focuses on approximating the model's behavior in the local region of a specific input instance. LIME provides interpretable insights into complex model predictions by implementing a simple surrogate model trained on a small number of data samples[62].

The LIME methodology involves generating perturbed samples x_i' by applying random perturbations to the original input data x. For each perturbed sample, the model's prediction $f(\mathrm{x}_i')$ is recorded. Subsequently, a simple, interpretable model g, such as a linear regression or decision tree, is trained on the perturbed data. The surrogate model approximates the complex model's behavior in the vicinity of the input instance by minimizing the following loss function:

$$\mathcal{L}(g) = \sum_i \pi_{\mathrm{x}}(\mathrm{x}_i') \cdot |f(\mathrm{x}_i') - g(\mathrm{x}_i')|^2 \tag{26}$$

Here, $\pi_{\mathrm{x}}(\mathrm{x}_i')$ is a locality kernel function that assigns higher weights to perturbed samples closer to the original instance x. The trained surrogate model effectively approximates the original model's behavior in the local neighborhood, providing interpretable insights into feature contributions and model decisions [63].

By combining Grad-CAM and LIME, we ensure both global and local interpretability in our ResNet-based framework for brain tumor detection, offering valuable insights for clinical decision-making and enhancing trust in the model's predictions.

Table 1 Performance Metrics achieved for Models on Dataset 1

Model	T-L (%)	T-Acc (%)	T-Prec (%)	T-Rec (%)	G-F1	NT-F1	M-F1	P-F1
M1	63.17	89.12	89.34	88.91	0.86	0.92	0.84	0.95
M2	70.63	89.83	90.00	89.50	0.88	0.93	0.87	0.93
M3	56.94	90.59	90.58	90.44	0.88	0.92	0.85	0.97
M4	39.75	92.26	92.40	92.26	0.91	0.97	0.85	0.97
M5	17.16	96.81	96.81	96.81	0.95	0.99	0.94	0.99
M6	16.72	96.21	96.35	96.05	0.96	0.97	0.96	0.97

M1: CNN, M2: VGG16, M3: MobileNetV2, M4: EfficientNetV2S, M5: DenseNet121, M6: ResNet

[1] G stands for Glioma,

[2] NT stands for No Tumor,

[3] M stands for Meningioma,

[4] P stands for Pituitary

[5] T stands for Test

Table 2 Performance Metrics achieved for Models on Dataset 2

Model	T-L (%)	T-Acc (%)	T-Prec (%)	T-Rec (%)	G-F1	NT-F1	M-F1	P-F1
M1	46.38	95.17	95.17	95.17	0.94	0.97	0.91	0.98
M2	7.51	98.15	98.29	98.08	0.99	0.99	0.97	0.98
M3	31.77	94.17	94.37	94.17	0.92	0.98	0.89	0.97
M4	14.75	97.30	97.30	97.30	0.96	0.99	0.95	0.98
M5	73.03	81.17	81.50	81.09	0.80	0.98	0.72	0.70
M6	4.11	99.36	99.36	99.36	0.99	1.00	0.99	0.99

M1: CNN, M2: VGG16, M3: MobileNetV2, M4: EfficientNetV2S, M5: DenseNet121, M6: ResNet

[1]G stands for Glioma,

[2]NT stands for No Tumor,

[3]M stands for Meningioma,

[4]P stands for Pituitary

[5]T stands for Test

Table 3 Performance Metrics achieved for Models on Dataset 3

Model	T-L (%)	T-Acc (%)	T-Prec (%)	T-Rec (%)	G-F1	NT-F1	M-F1	P-F1
M1	54.23	94.87	94.56	94.61	0.94	0.95	0.92	0.97
M2	5.12	98.91	98.98	98.91	0.99	1.00	0.99	0.99
M3	34.26	94.06	94.06	94.06	0.94	0.96	0.88	0.97
M4	11.24	97.81	97.88	97.81	0.97	0.99	0.95	0.99
M5	45.61	88.46	88.77	88.32	0.82	0.99	0.79	0.90
M6	5.12	98.91	98.98	98.91	0.99	1.00	0.99	0.99

M1: CNN, M2: VGG16, M3: MobileNetV2, M4: EfficientNetV2S, M5: DenseNet121, M6: ResNet

[1]G stands for Glioma,

[2]NT stands for No Tumor,

[3]M stands for Meningioma,

[4]P stands for Pituitary

[5]T stands for Test

3.2.1 Formula for local surrogate model

The local surrogate model g is trained by solving the following optimization problem:

$$\hat{g} = \arg\min_{g} \sum_{i} \pi_{\mathrm{x}}(\mathrm{x}'_i) \cdot |f(\mathrm{x}'_i) - g(\mathrm{x}'_i)|^2$$

where: - $f(\mathrm{x}'_i)$ represents the prediction from the original model for perturbed instance x'_i, - $g(\mathrm{x}'_i)$ is the prediction from the surrogate model g, - $\pi_{\mathrm{x}}(\mathrm{x}'_i)$ is a weight function that assigns higher importance to perturbed samples closer to the original instance x.

4 Results

This section presents the evaluation metrics for various models trained and tested on three datasets. The metrics include test loss, accuracy, precision, recall, and F1-scores for different tumor classes (Glioma, No Tumor/Healthy, Meningioma, and Pituitary). The proposed model is generalized on 3 different datasets, and the quantitative results are presented in Table 1 for Dataset 1, Table 2 for Dataset 2 and Table 3 for Dataset 3.

4.1 Results of models trained on different datasets

The performance metrics on Dataset 1, presented in Table 1, show that DenseNet-121 and ResNet achieved higher results, with accuracies of 96.81% and 96.21%, respectively, and precision and recall above 96%, as well as F1-scores close to 0.99 for key classes. EfficientNetV2S followed with 92.26% accuracy, balanced precision(92.40%) and recall (92.26%). MobileNetV2 performed fairly with 90.59% accuracy and lower F1-scores (0.85–0.97). VGG16 and the custom CNN recorded the lowest performance, with an accuracy below 90%, higher loss (70.63% and 63.17%), and lower F1-scores (0.84–0.93), indicating less effectiveness in tumor detection.

The performance analysis of Dataset 2 presented in Table 2 presents significant findings. The ResNet model outperformed others, achieving the lowest test loss (4.11%) and the highest test accuracy (99.36%), along with nearly perfect precision and recall (99.36% each). F1 scores for glioma, no tumor, meningioma, and pituitary tumors were exceptionally high, highlighting its robustness. VGG16 also showed strong results with a test loss of 7.51% and an accuracy of 98.15%. Its F1 scores ranged from 0.98 to 0.99 across tumor categories, demonstrating its reliability. EfficientNetV2S had decent performance (test loss: 14.75%, accuracy: 97.30%), displaying

Table 4 Ablation study for Proposed Model

Model	Accuracy (%)	Precision (%)	Recall (%)
R	89.62	91.44	87.28
R - ULL (Unfreeze Last 10 Layers)	96.22	97.29	95.15
R - ULL + SE (Unfreeze Last 10 Layers + SE)	98.62	98.74	97.28
R - ULL + SE + GAP (Global Average Pooling)	98.69	98.94	98.19
R - ULL + SE + GAP + DO (Dropout)	99.13	98.94	98.03
R - ULL + SE + GAP + DO + D (Dense)	99.11	99.14	99.28
R - ULL + SE + GAP + DO + D + AO (Adam Optimizer)	98.58	98.58	98.54
R - ULL + SE + GAP + DO + D + AO + LRP (LR Plateau)	99.86	99.86	99.87

[1]ULL stands for Unfreeze Last 10 Layers

[2]SE stands for Squeeze and Excitation

[3]GAP stands for Global Average Pooling

[4]DO stands for Dropout

[5]D stands for Dense Layer

[6]AO stands for Adam Optimizer

[7]LRP stands for Learning Rate Plateau

[8]R stands for ResNet

high precision and recall. MobileNetV2 recorded moderate results (test loss: 31.77%, accuracy: 94.17%), while DenseNet121 struggled with the highest test loss (73.03%) and lowest accuracy (81.17%). The custom CNN model also fell short (test loss: 46.38%, accuracy: 95.17%).

The performance of various models on Dataset 3 highlights the superior capability of ResNet. It achieves a test accuracy of 98.91%, a test precision of 98.98%, and a recall of 98.91%. The class-wise F1 scores for glioma, no tumor, meningioma, and pituitary are all close to or equal to 0.99, unprecedented classification performance across all categories.VGG16 also illustrates higher performance, similar to ResNet in accuracy (98.91%) and test precision (98.98%), with strong class-wise F1 scores. It shows high stability in classifying all tumor types and healthy cases, making it a better alternative to ResNet. EfficientNetV2S performs slightly lower with a test accuracy of 97.81%, but still maintains high precision and recall values, describing its effectiveness for this task. MobileNetV2, performs reasonably well with a test accuracy of 94.06% but shows a slight decline in precision and recall compared to the top models. While it is efficient and lightweight, it is not as strong in complex classification scenarios. DenseNet121 records the lowest accuracy of 88.46%, with notable drops in glioma and meningioma classification (F1 scores of 0.82 and 0.90, respectively), indicating challenges in capturing the complex features required for accurate classification.

4.2 Results for the proposed model

The modifications made to ResNet reflect progressive improvements in all cases; the base model reports a score of 89.62%, and unfreezing the last 10 layers increases it to 96.22%. Further improvements above 99% have been achieved through the incorporation of Squeeze and Excitation, Global Average Pooling, Dropout and Dense features. The introduction of the Adam optimizer does cause a slight decrease, but the final model with a Learning Rate Plateau achieves near-perfect metrics: 99.86% accuracy, 99.86% precision and 99.87% recall. The

Table 5 Performance Metrics for Various Dynamic Learning Rate Modifiers on Proposed Model (ResNet)

Model	T Accuracy (%)	T Precision (%)	T Recall (%)	G F1	NT F1	M F1	P F1
LR Plateau	99.86	99.86	99.86	99.84	99.87	99.85	99.86
Adam	98.58	98.58	98.54	99.22	99.25	97.04	98.71
RMSProp	98.79	98.82	98.70	98.09	99.87	97.71	99.27
AMSGrad	95.59	95.92	95.91	98.45	94.06	92.08	98.38

[1]G stands for Glioma,

[2]NT stands for No Tumor,

[3]M stands for Meningioma,

[4]P stands for Pituitary

[5]T stands for Test

Table 6 Performance Metrics for Proposed Model Across Three Datasets

Dataset	T Accuracy (%)	T Precision (%)	T Recall (%)	G F1	NT F1	M F1	P F1
Dataset 1	97.27	0.968	0.958	0.963	0.980	0.962	0.991
Dataset 2	99.65	0.997	0.994	0.996	0.997	0.994	0.999
Dataset 3	99.22	0.994	0.986	0.990	0.997	0.984	0.996

[1]G stands for Glioma,

[2]NT stands for No Tumor,

[3]M stands for Meningioma,

[4]P stands for Pituitary

[5]T stands for Test

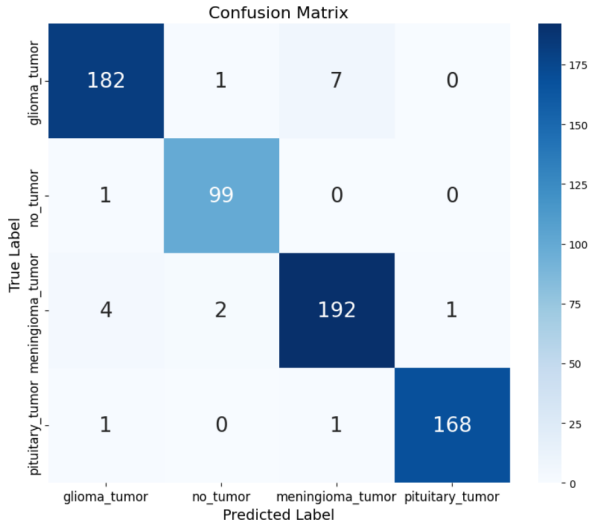

Fig. 5 Confusion Matrix for Dataset 1

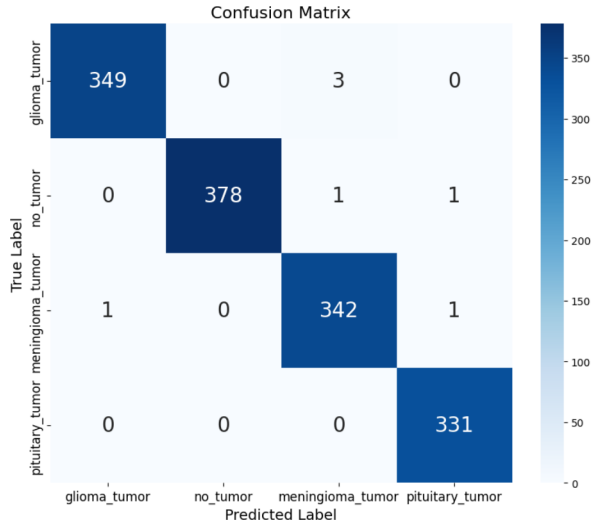

Fig. 7 Confusion Matrix for Dataset 2

ablation study conducted to achieve the proposed model is presented in Table 4.

The LR Plateau dynamic learning rate modifier performed best across all metrics. It obtained a test accuracy of 99.86%, with high precision and recall values averaging 99.86%. Class-specific F1-scores were equally impressive, with Glioma (G), No Tumor (NT), Meningioma (M), and Pituitary (P) F1-scores of 99.84%, 99.87%, 99.85%, and 99.86%, respectively. These results indicate that the LR Plateau approach effectively balances precision and recall while maintaining consistency in performance across all tumor types. In comparison, other methods such as Adam, RMSProp, and AMSGrad achieved lower accuracy and F1-scores as presented in Table 5, highlighting the superiority of the LR Plateau method.

The results underscore the transformative potential of deep learning in medical diagnostics, particularly brain tumor classification. ResNet, with accuracies reaching **99.65%**, demonstrated unparalleled performance across all tumor types, benefiting from advanced feature extraction enabled by residual connections and SE blocks. These findings establish a benchmark for integrating AI into clinical workflows, providing precise, interpretable, and reliable diagnostic support for neurological oncology.

The proposed model demonstrates excellent performance across all three datasets, as shown in Table. 6. On Dataset 1, it achieves a test accuracy of 97.27% with a precision of 0.968 and recall of 0.958, along with high F1 scores for all classes, indicating better generalization and classification. For Dataset 2, the model's performance improves further, attaining a test accuracy of 99.65%, precision of 0.997, and recall of 0.994, with nearly perfect F1 scores across all tumor categories. On Dataset 3, the model maintains an accuracy of 99.22%, with consistent precision and recall values. The F1 scores remain close to 1 for all classes, showcasing the robustness and reliability of the proposed model in effectively differentiating between glioma, no tumor, meningioma, and pituitary tumors across varying datasets.

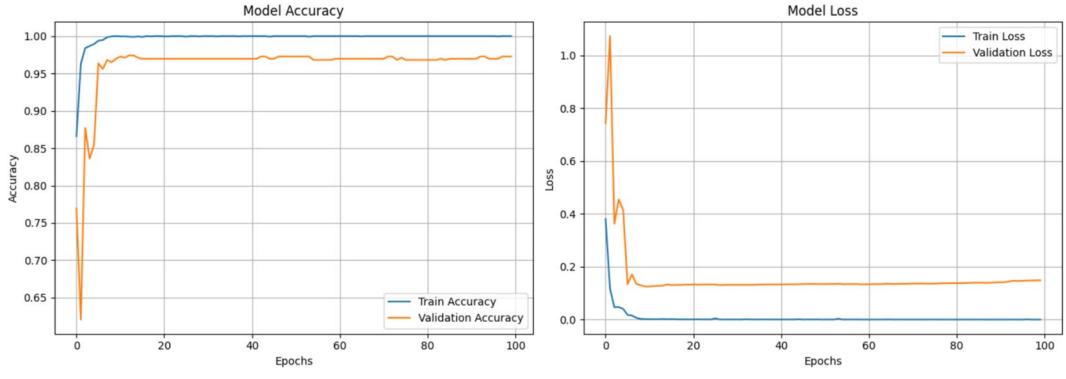

Fig. 6 Accuracy and Loss Plots of Dataset 1 for Enhanced ResNet

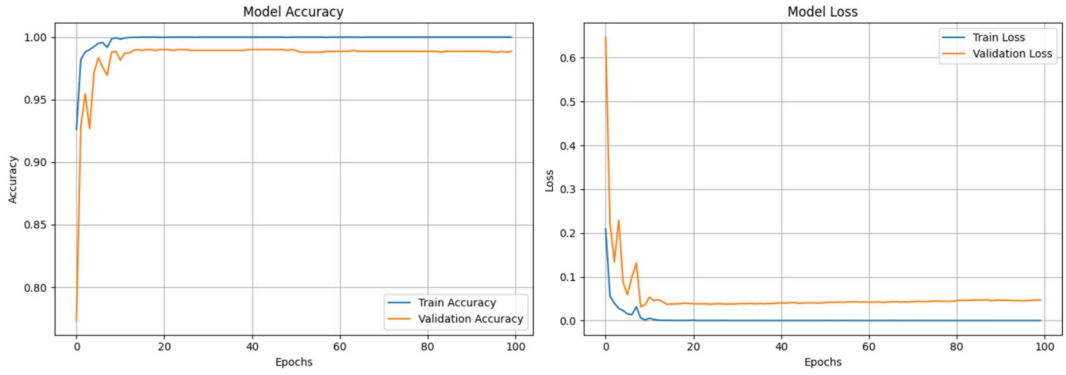

Fig. 8 Accuracy and Loss Plots of Dataset 2 for Enhanced ResNet

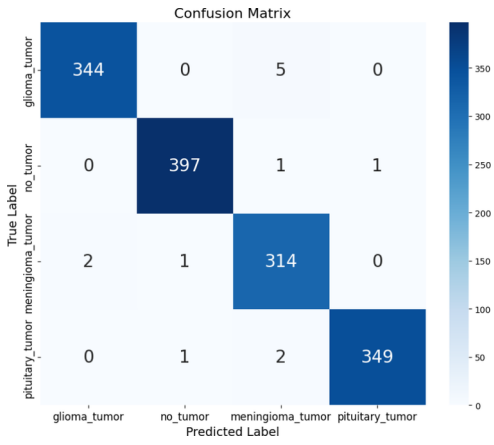

Fig. 9 Confusion Matrix for Dataset 3

4.3 Confusion metrics and plots for the proposed model

The confusion matrix in Fig. 5 shows high classification performance, with 182/190 correct predictions for glioma (95.8%), 99/100 for no tumor (99%), 192/199 for meningioma (96.5%), and 168/170 for pituitary tumors (98.8%). Misclassifications were minimal, the highest being 7 cases predicted as meningioma. The training and validation plots in Fig. 6 show rapid convergence, with validation accuracy reaching approximately 96% within 10 epochs and remaining stable. The training loss dropped below 0.05, and the validation loss stabilized around 0.15, indicating effective learning and minimal overfitting.

The confusion matrix in Fig. 7 supports these results, indicating minimal misclassifications, particularly for glioma, no tumor, and pituitary cases. Accuracy and loss plots in Fig. 8 confirm that Enhanced ResNet converges quickly and maintains consistent performance throughout 100 epochs, depicting it as the top model for Dataset 2, followed by VGG16 and EfficientNetV2S.

The confusion matrix presented in Fig. 9 confirms that misclassifications are minimal, making ResNet the most reliable model among those evaluated. The accuracy and loss plots for Fig. 10 reveal rapid convergence within the first 20 epochs, with both training and validation accuracy reaching stability of nearly 99%. The loss curves also slightly decline at lower values, confirming the model's strong learning ability and robustness on Dataset 3.

The ResNet model consistently outperformed other architectures across all datasets, achieving the highest test accuracy ranging from **99.36% to 99.65%**. It demonstrated superior precision, recall, and F1-scores, particularly excelling in classifying glioma and pituitary tumors. Integrating residual connections and Squeeze-and-Excitation (SE) blocks played a critical role in improving feature extraction and reducing overfitting.

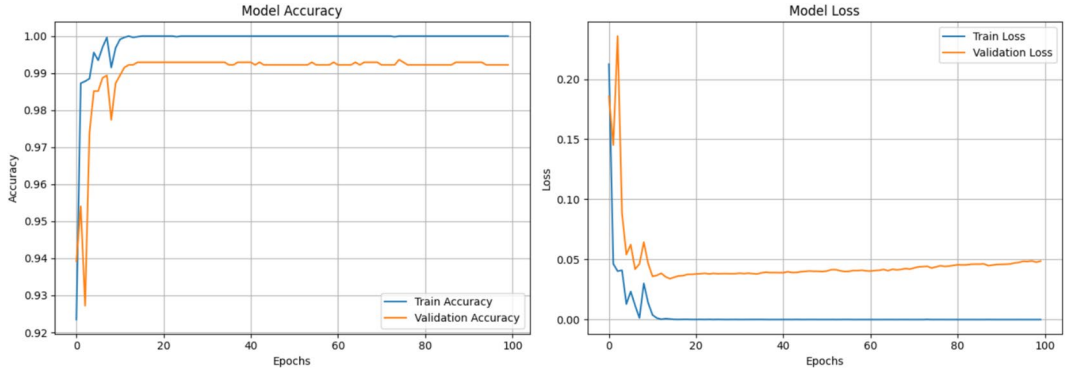

Fig. 10 Accuracy and Loss Plots of Dataset 3 for Enhanced ResNet

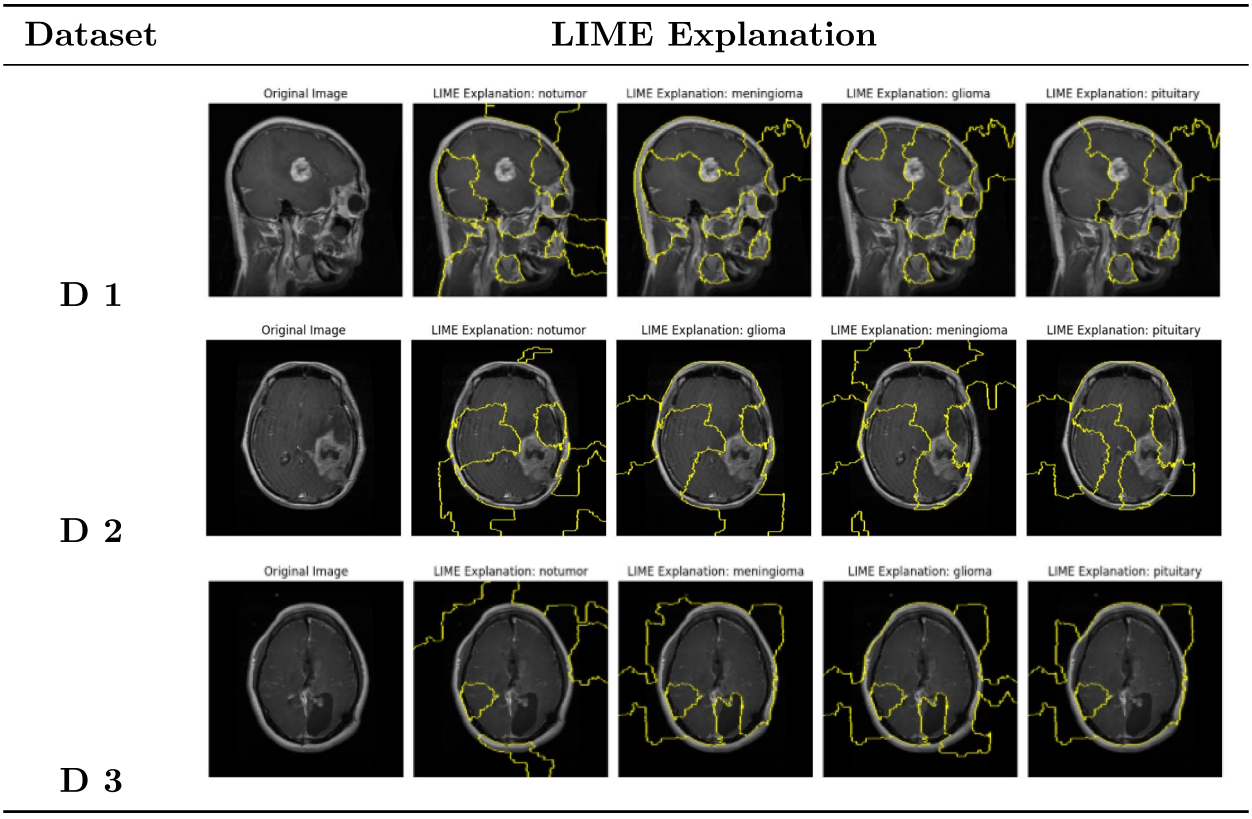

Fig. 11 LIME Explainability for Brain Tumor Detection Across Three Datasets

4.4 LIME and Grad-CAM implementation on Proposed Model

The LIME-based explainability results illustrated in Figure. 11 demonstrate the model's focus areas across three different datasets for brain tumor detection. For each dataset, LIME highlights regions of the MRI scans that influence the classification of no tumor, meningioma, glioma, and pituitary tumors. The visualizations show that while LIME effectively identifies certain relevant regions, the highlighted superpixels often appear to overlap across different tumor types and healthy scans. This overlap indicates that LIME struggles to distinguish precise, class-specific features, making it challenging to interpret model decisions with clinical clarity.

In the first dataset, although some focused regions align with the expected tumor locations, the highlighted area s are not sharply defined and appear scattered. Similar patterns are observed in datasets 2 and 3, with LIME outputs showing diffuse, broad regions of interest rather than clear tumor boundaries. This lack of granularity reduces the effectiveness of LIME in isolating subtle differences between tumor types, which is critical in medical diagnostics.

The superpixel approach of LIME is beneficial in general domains, but it does not offer the level of detailed, localized explanation required for MRI-based tumor classification. The highlighted regions are too coarse and do not correspond to well-defined anatomical or pathological structures, reducing the practical utility of these explanations for medical professionals. As a result, although LIME provides a surface-level understanding of the model's focus areas, it gives less precise, actionable insights necessary for clinical interpretation.

For more meaningful and clinically relevant visual explanations, techniques such as Grad-CAM, which inherently utilize the neural network architecture to provide spatial precise and class-discriminative heatmaps, may be better suited. In conclusion, while LIME contributes to basic interpretability, its current implementations can be improved and enhanced in the future to meet the standards demanded in medical imaging applications.

The Grad-CAM visualizations are depicted in Figure. 12 offer clear spatial insights into the model's decision-making process across glioma, meningioma, and pituitary tumor classes for three datasets. For glioma, the heatmaps prominently highlight irregular and diffused tumor regions, aligning well with the clinical characteristics of this tumor type. In the case of meningioma, the activations are finely focused around well-defined tumor boundaries, indicating the model's ability to capture distinct and localized tumor structures. For pituitary tumors, the visualizations emphasize the regions

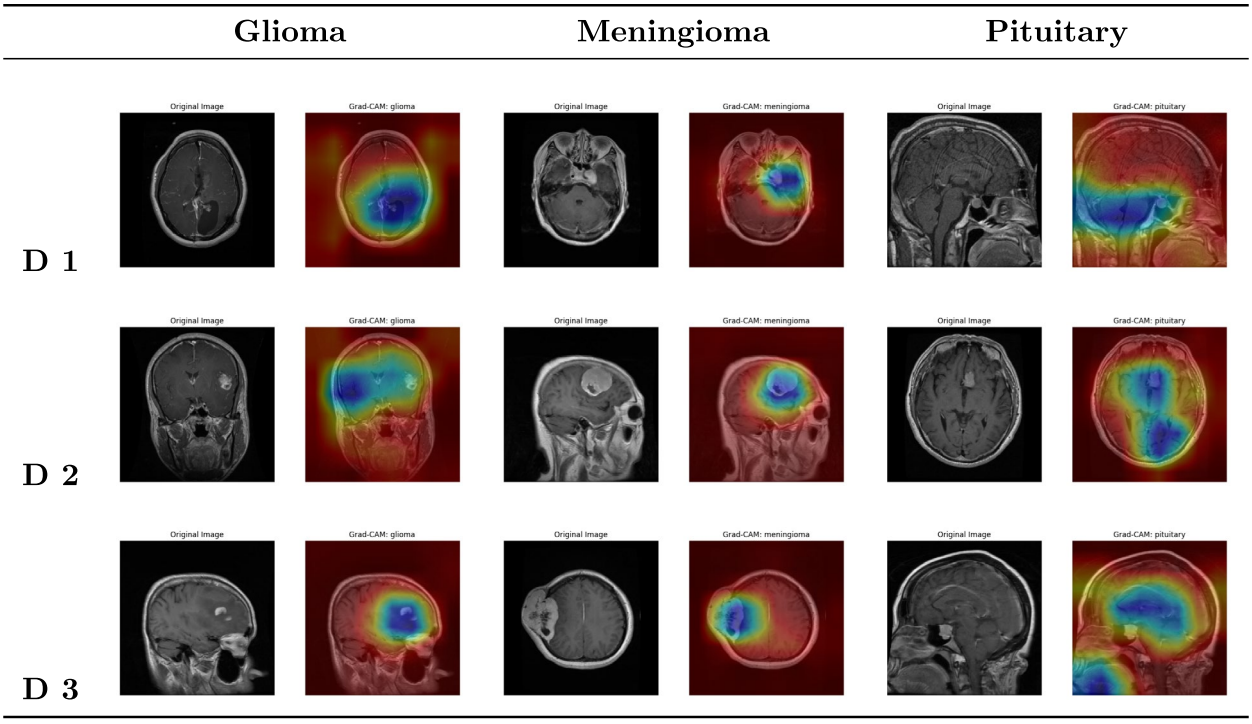

Fig. 12 GradCAM Visualizations for Glioma, Meningioma, and Pituitary Classes across three datasets

near the central part of the brain, along with the tumorous regions, depicting the model's understanding of the typical non-anatomical location of these tumors. Across all datasets, these Grad-CAM maps show consistent activations that correlate with clinically relevant features except for the pituitary tumors, confirming the reliability and interpretability of the model's predictions. When LIME is considered a more generic explainable method, Grad-CAM provides high-resolution focus on tumor-specific areas, providing confidence that the classification decisions are grounded in meaningful and medically significant regions.

5 Discussion

The proposed SE-ResNet framework demonstrates outstanding performance in brain tumor classification, achieving accuracies exceeding 99% across three independent MRI datasets. The ablation study (Table 4) shows that each architectural refinement contributed incrementally to performance. Unfreezing the last ten layers improved transfer learning effectiveness, while the integration of Squeeze-and-Excitation (SE) blocks enhanced channel-wise feature representation, enabling the model to focus on tumor-specific cues. Global Average Pooling reduced parameter redundancy, Dropout alleviated overfitting and enriched feature interactions with additional dense layers. The incorporation of an adaptive learning rate strategy (LR Plateau) further stabilized training to ensure consistent gains across datasets. Together, these modifications elevated the baseline ResNet accuracy to near-perfect levels. The strong performance across multiple datasets highlights the robustness and generalization of the proposed framework, distinguishing it from studies limited to single-dataset validation. Grad-CAM visualizations confirmed that the model consistently attended to clinically relevant tumor regions, reflecting structural features commonly identified by radiologists. LIME provided complementary insights into feature importance, though its coarser spatial resolution compared to Grad-CAM suggests gradient-based methods may be more effective for localized medical interpretability. The combination of performance and explainability demonstrate the potential of SE-ResNet as a clinically trustworthy decision-support system.

At the same time, several limitations must be acknowledged. Although the datasets used are more diverse than in many prior works, they still represent curated MRI collections with four classes, which may not capture rare tumor subtypes or inter-institutional variability in imaging protocols. Misclassifications, while minimal, were most notable in meningioma cases, indicating that irregular tumor boundaries remain challenging. The reliance on 2D MRI slices resized to 250×250 pixels introduce a trade-off between computational efficiency and the preservation of fine-grained anatomical detail: future extensions to 3D volumetric scans could provide richer contextual information and improve tumor delineation. Additionally, fine-tuning required careful selection of

layers and learning rates, which remains computationally intensive and sensitive to hyperparameters. The reliance on high-performance GPUs such as NVIDIA A100 may also limit accessibility in resource-constrained settings. Finally, while Grad-CAM and LIME enhance transparency, they provide post-hoc explanations that are inherently qualitative and require clinical validation to ensure interpretability in real-world practice.

It is also essential to consider the scope of this framework beyond MRI. While SE-ResNet proved highly effective for structural MRI, its direct application to modalities such as positron emission tomography (PET) are less straightforward. MRI provides high spatial resolution and strong tissue contrast, whereas PET has lower resolution and higher noise, while measuring functional rather than structural information. Adapting this framework for PET would therefore require substantial modifications, including harmonization of resolution, denoising, and normalization in preprocessing, as well as adjustments to feature extraction layers to capture metabolic rather than anatomical patterns. This highlights that although the overall strategy is extensible, modality-specific adaptations are essential for reliable performance across different imaging domains.

6 Conclusion

The study describes an improved systematic enhancement of the SE-ResNet model to brain tumor classification that reaches close to perfect accuracy in three different MRI data sets. Through the successive stages of refined architecture, including unfreezing of deeper layers, adding Squeeze-and-Excitation blocks, global average pooling, dropout, and dense layers, the model proved to have an outstanding representation of features and overall classification accuracy of above 99%. The combination of explainable AI techniques and the use of Grad-CAM ensured the predictions were based on clinically meaningful tumor areas, improving the interpretability and confidence in the system. High predictive accuracy and open-minded decision-making is the synergy that makes the model potentially an important clinical decision support tool to reinforce confidence in its deployment to help radiologists.

Based on such positive findings, further studies will aim at generalizing the framework to broader clinical applicability and strength. This also involves training bigger and more heterogeneous datasets to capture rare tumor types and different radiologic acquisitions and expand the model to discuss 3D volumetric scans, which provides more detailed spatial information. Integration Multi-modal integration, like the integration of MRI with CT or PET can further enhance the representation of the features. The SE-ResNet framework can be further adapted to classify tumor subtypes and capture more subtle intra-tumoral characteristics. Achieving this will require more granular datasets with detailed subtype annotations, along with potential architectural enhancements, such as multi-scale feature extraction and attention mechanisms that focus on finer patterns. Incorporating advanced interpretability techniques beyond Grad-CAM and LIME will provide more precise and actionable explanations of model predictions. Ultimately, the goal is to deploy this high-performing, interpretable AI model into clinical workflows to assist the radiologist's efficiency and support more confident, data-driven decision-making.

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Author contributions
K.A. and A.C.G. conceptualized the study, designed the methodology, and con- ducted the experiments. K.A. drafted the initial manuscript. T.S. and R.K. performed result analysis and validation. T.S., G.N., and R.K. supervised the research. R.K., a consultant in radiation oncology, conducted a clinical assessment of the brain tumor detection from MRI images to ensure accuracy and diagnostic usability. His evaluation confirms the fidelity and applicability of the proposed model for medical applications. All authors have read and approved the final manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable

Consent for publication
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Competing interests
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